

Omar Hafez Khalil

Artificial Intelligence Engineer

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Professional Summary

AI Engineer specializing in Machine Learning, Deep Learning, Computer Vision, and Generative AI, with hands-on experience designing and deploying end-to-end AI systems. Experienced in developing production-oriented Computer Vision pipelines, Retrieval-Augmented Generation (RAG) applications, and multi-agent AI workflows. Skilled in PyTorch, FastAPI, LangChain, LangGraph, LLMs, vector databases, hybrid retrieval, and local LLM deployment, with a strong focus on building practical, scalable, and deployable AI solutions.

Technical Skills

Programming Languages: Python | JavaScript | SQL | C++ | Java

Machine Learning & Deep Learning: PyTorch | Scikit-learn | Neural Networks | Transfer Learning | Fine-Tuning | Hyperparameter Tuning | Model Evaluation | Feature Engineering

Computer Vision: OpenCV | Object Detection | Image Classification | OCR | Ultralytics (YOLOv8) | ResNet-50

Generative AI & LLM Engineering: LLMs | RAG | AI Agents | Model Context Protocol (MCP) | LangChain | LangGraph | Transformers | Embeddings | Prompt Engineering | Vector Databases (ChromaDB) | Cloud & Local LLMs (Gemini, ChatGPT, Grok, Ollama)

Model Serving & APIs: FastAPI | REST APIs | AI inference endpoints | Asynchronous APIs

Backend & Databases: Node.js | Express.js | MongoDB | Mongoose | Database Design

Developer Tools: Git | GitHub | Postman | Jupyter Notebook | Google Colab

Projects

Venezia Bank – AI-Powered Digital Banking System | *AI, Flutter, Laravel, MySQL, FastAPI*

- Developed a comprehensive AI-powered digital banking ecosystem featuring a cross-platform mobile application for customers and a web-based platform for bank employees, supported by a Laravel backend and integrated AI services. The platform provides core banking services including payments, transfers, account management, loans, investments, financial reporting, and fraud monitoring.
- Served as an AI Engineer responsible for designing and implementing 4 core AI modules, contributing to an ecosystem integrating 8+ AI capabilities, including financial fraud detection, receipt recognition, customer churn prediction, product recommendation, stock prediction, loan decision support, employee decision support and an LLM/RAG financial chatbot.
- Designed and deployed AI services as FastAPI-based microservices, enabling integration between AI models and the core banking platform through RESTful APIs.
- 🐙 **GitHub Repository**

Receipt Recognition System | *FastAPI, Ultralytics(YOLOv8), PyTorch, ResNet-50, Gemini API, OpenCV*

- Developed an end-to-end AI pipeline converting multilingual receipt images into structured financial data through 4 automated stages: validation, region detection, OCR, and product classification.
- Achieved 98.83% validation accuracy, 0.7693 mAP@50, and 0.93 F1-score across 15 product categories.
- Deployed the pipeline as a FastAPI REST service with 12 asynchronous workers, supporting Arabic and English receipts.
- 🐙 **GitHub Repository**

DocMind AI (Multi-Agent) | *Python, LangChain, LangGraph, ChromaDB, Hybrid Retrieval, Ollama, Streamlit*

- Built a custom multi-agent document intelligence pipeline orchestrating 4 specialized agents for relevance checking, extraction, reasoning, and response generation.
- Implemented hybrid retrieval, persistent ChromaDB storage, and document caching for reusable local knowledge.

- Developed an offline-first document analysis application supporting PDF, DOCX, TXT, and Markdown using local Ollama LLMs.
-  **GitHub Repository**

Banking Finance Chatbot (RAG) | *FastAPI, LangChain, ChromaDB, Ollama, Gemini API, Embeddings*

- Developed a domain-specific RAG chatbot for banking and financial assistance using vector retrieval and LLM-based response generation.
- Integrated ChromaDB, embeddings, and local/cloud LLM inference through Ollama and Gemini API.
- Achieved 88.29% BERTScore in response evaluation and deployed the application as a FastAPI service.
-  **GitHub Repository**

AI-powered Banking Retention & Product Recommendation System | *FastAPI, XGBoost, LightGBM*

- Developed an intelligent banking retention system that predicts customer churn and automatically recommends personalized financial products through a real-time REST API. Implemented a dual-machine-learning pipeline for prediction and recommendation.
- Achieved 0.88 F1-score for both churn prediction and product recommendation.
-  **GitHub Repository** |  **Live Demo**

NutriVision: Food Classification | *PyTorch, ResNet50, EfficientNet-B0, TensorBoard, TorchMetrics*

- Developed a deep learning food classification system covering 30 categories, comparing custom CNN and transfer learning architectures.
- Evaluated Custom CNN, ResNet-50, and EfficientNet-B0 models and deployed an interactive inference application.
-  **GitHub Repository** |  **Live Demo**

Education

Arab International University (AIU)

2021 – 2026

- Bachelor's Degree in Informatics & Communications Engineering
- Artificial Intelligence Engineering
- GPA: 3.36/4.0

Courses

- **Agentic AI with LangChain and LangGraph** (Coursera, Completed Jun. 2026)
- **Machine Learning** (Coursera, Completed Mar. 2026)
- **Generative AI with Large Language Models** (Coursera, Completed Sep. 2025)
- **Deep Learning & Computer Vision**
 - **PyTorch for Deep Learning & Machine Learning** (freeCodeCamp, Jan. 2025 – Aug. 2025)
- **Machine Learning with Python and Scikit-Learn** (freeCodeCamp, Jul. 2024 – Dec. 2024)

Languages

- Arabic — Native
- English — Professional Working Proficiency